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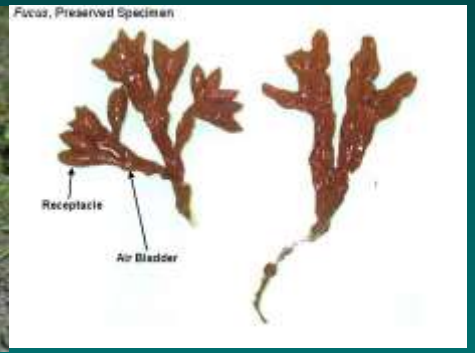
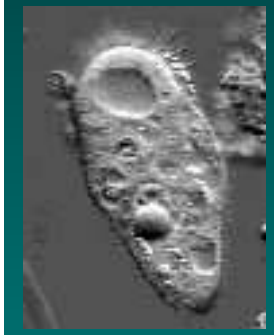
Kingdom: Protista





"The Protists

Prepared by

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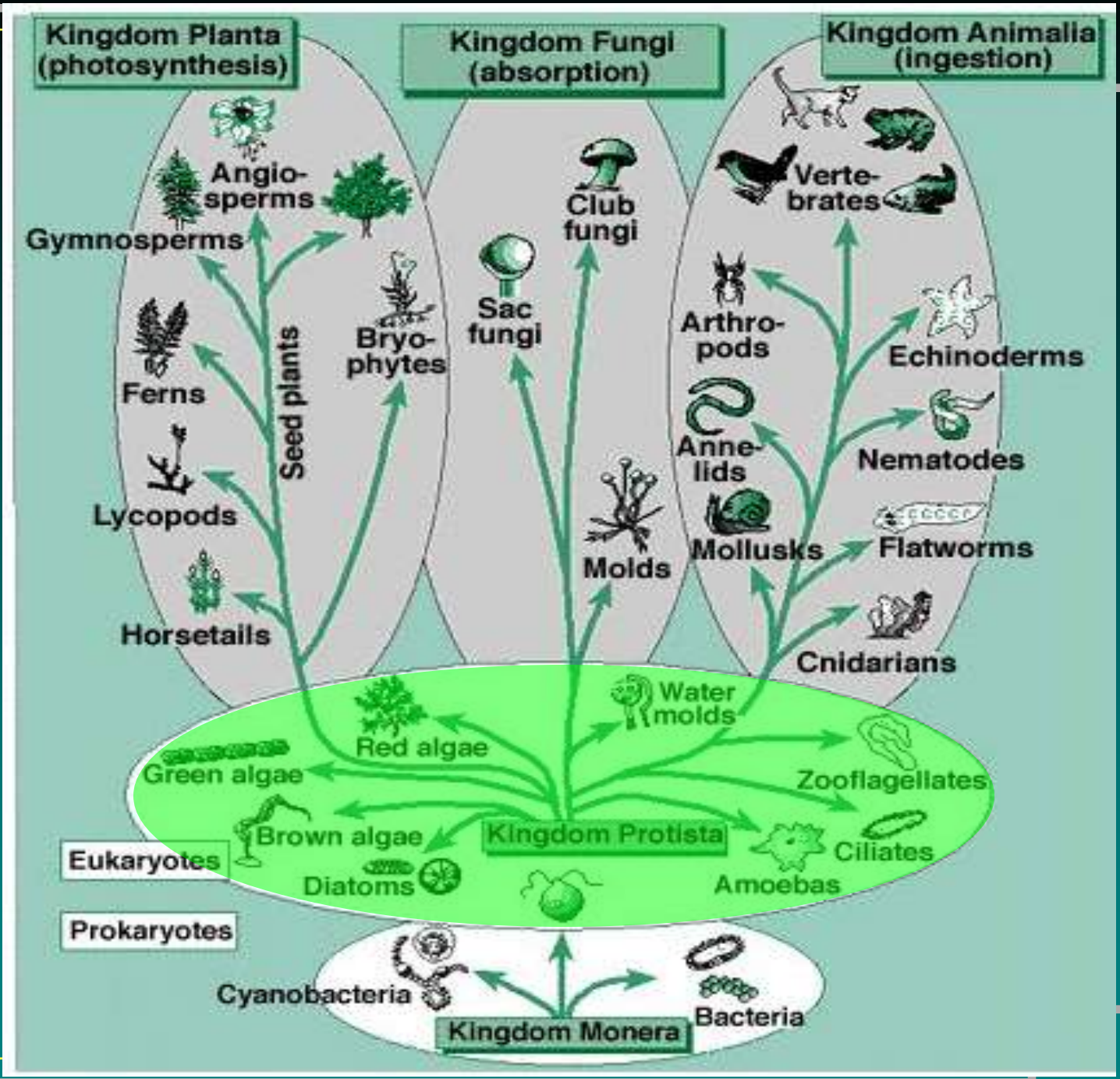
Department of Botany and Microbiology, Faculty of Science, Minia University, Egypt

Animal_like?!

Plant-Like Or Fungus-like?!

The Five Kingdoms



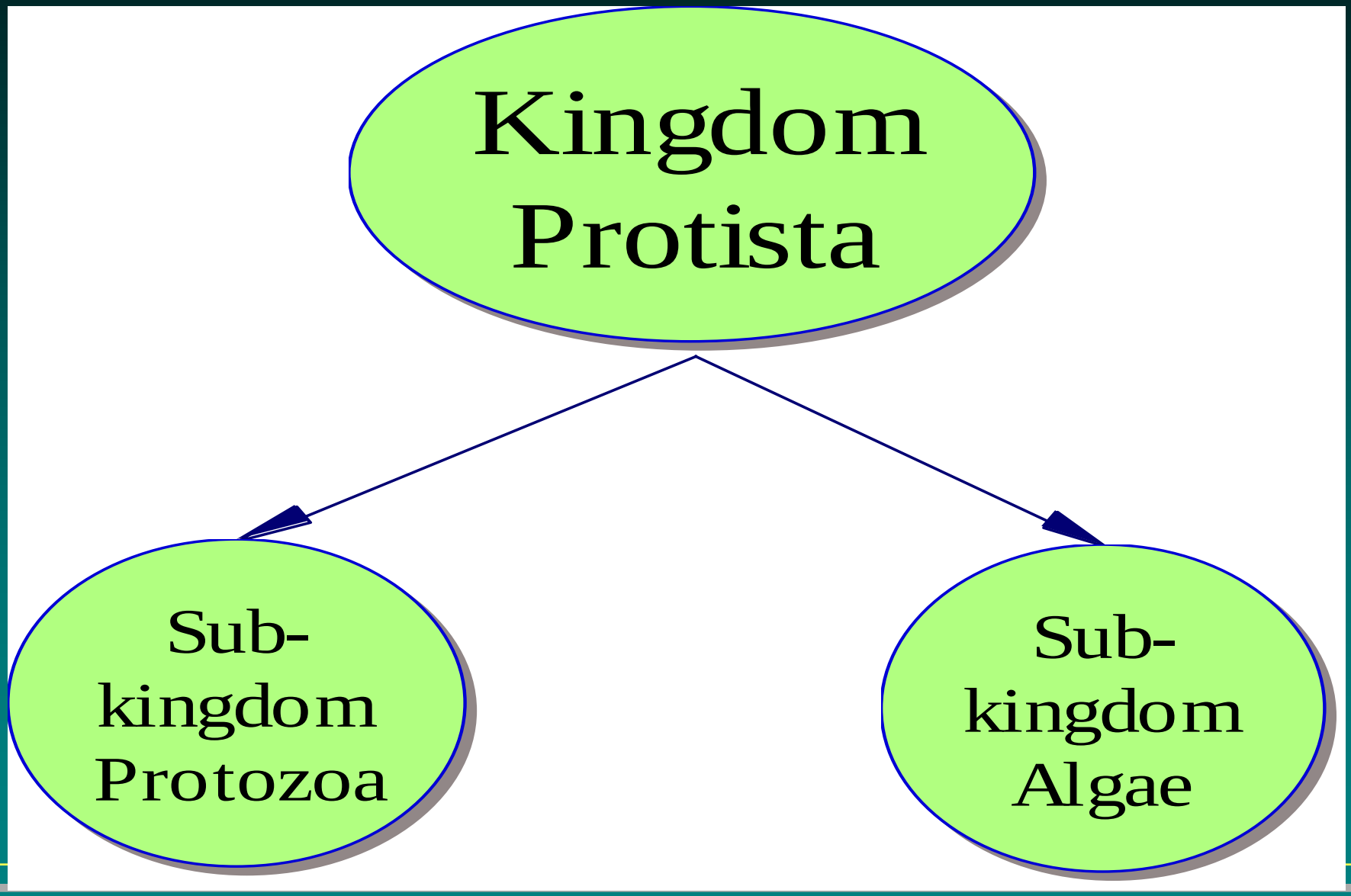
Commonalities / Differences in the Pr otist Kingdom

- **All are eukaryotes (cells with nuclei).**
- **Live in moist surroundings.**
- **Unicellular or multicellular.**
- **Autotrophs, heterotrophs, or both.**
- **Some can move - others cannot.**

Life Cycle

- **A few protists are entirely asexual.**
  - **binary fission** - equal halves
  - **budding** - progeny cell smaller
  - **schizogony** - multiple fission -into multiple individuals simultaneously
  - **Spores production** – zoospores or aplanospores
- **Cysts form at some point during the life cycle of many protists. Cysts are resistant cells that are capable of surviving harsh conditions.**
- **Most reproduce sexually by meiosis and then go on to reproduce asexually.**

Second classification



Kingdom Protista: Classification

- Sub-Kingdom Protozoa
  - Protozoans are animal-like protists. They are very sim ple heterotrophic organisms.
- Sub-Kingdom Algae
  - Algal protists are plant-like. They are very simple aut otrophic organisms.

Systematists <u>now</u> divide protists into as many as 2 0 separate kingdoms!!!.

Algae

- Plant-like protists are algae.
- Algae are eukaryotic autotrophs.
- They, along with other eukaryotic autotrophs, form t he foundation of Earth's food chains.
- They produce much of Earth's oxygen.
- Algae are grouped according to their color and bioch emical differences, such as the chemistry of the cell wall and the way they store reserve food.

Unicellular

- **There are three unicellular phyla of algae:**
  - Phylum Euglenophyta
  - Phylum Bacillariophyta
  - Phylum Dinoflagellata

Multicellular

- **Multicellular** algae are classified by color.

- **1. Rhodophyta** are red seaweeds.

- They are found in warm or cold marine environments along coast lines in deeper water.

- They absorb green, violet, and blue light waves. The se light waves are able to penetrate below 100 meters .

Rhodophyta

Porphyra

Gigarina

Filamentous Red Alga

Corraline Red Alga

- **2. Phylum Phaeophyta** is made up of the brown algae.

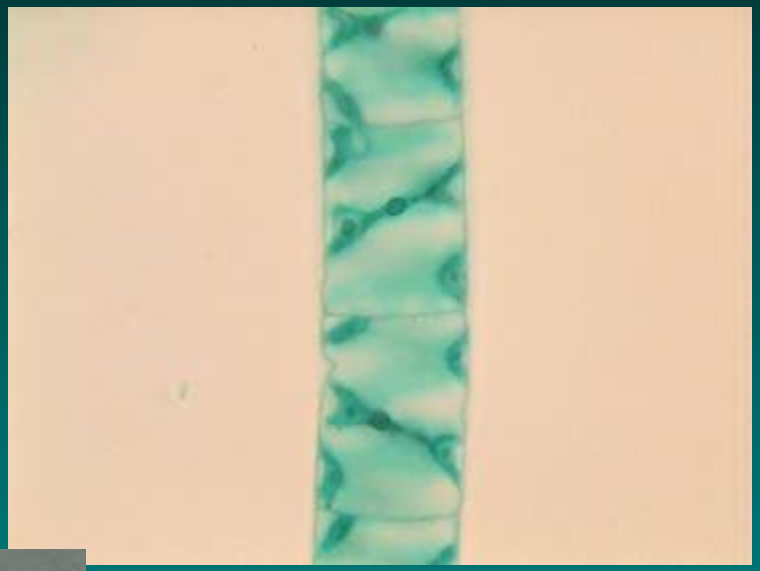
- They are found in cool saltwater along rocky coasts.

- Giant Kelp are the largest and most complex brown algae. They have hol d fasts and air bladders.

- **3. Phylum Chrysophyta** is made up of the the yellow green, golden browns and diatoms (now bacillariophyt a)

Dinobryon

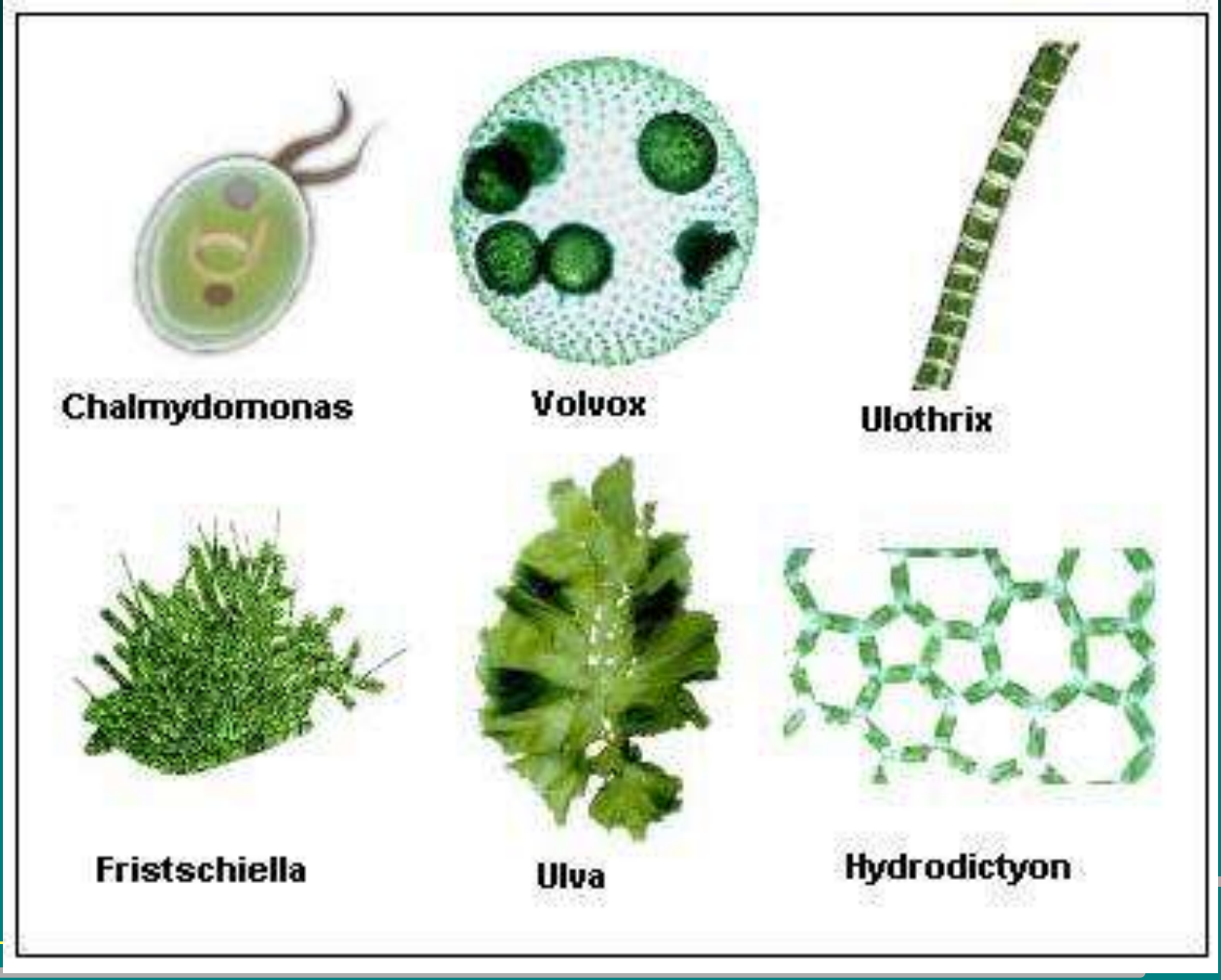
Diatom



Vaucheria

- **4. Phylum chlorophyta:** The last of the multicellular algae are the green algae from the **Phylum chlorophyta**.

Most green algae are found in fresh water habitats.



Chlamydomonas are actually unicellular and flagellated.

A Volvox is a hollow boll composed of hundreds of flagell
ted cells in a single layer.

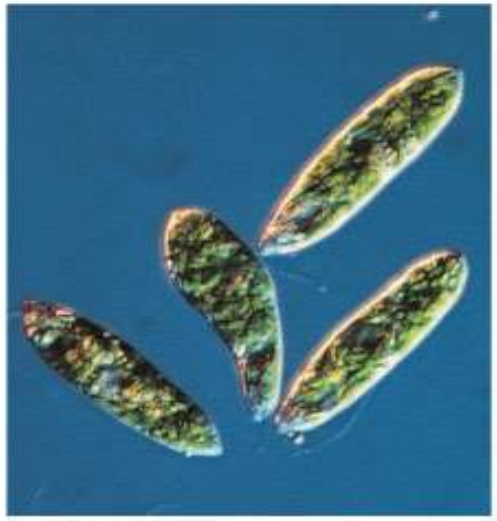
<u>Chara</u>- very closely related to first land plants

Unicellular Plant-Like Protists

- phytoplankton- small, photosynthetic organisms found near the surface of bodies of water
  - --contain chlorophyll/chloroplasts
  - --carry out about ½ of photosynthesis on Earth (produce lots of oxygen)

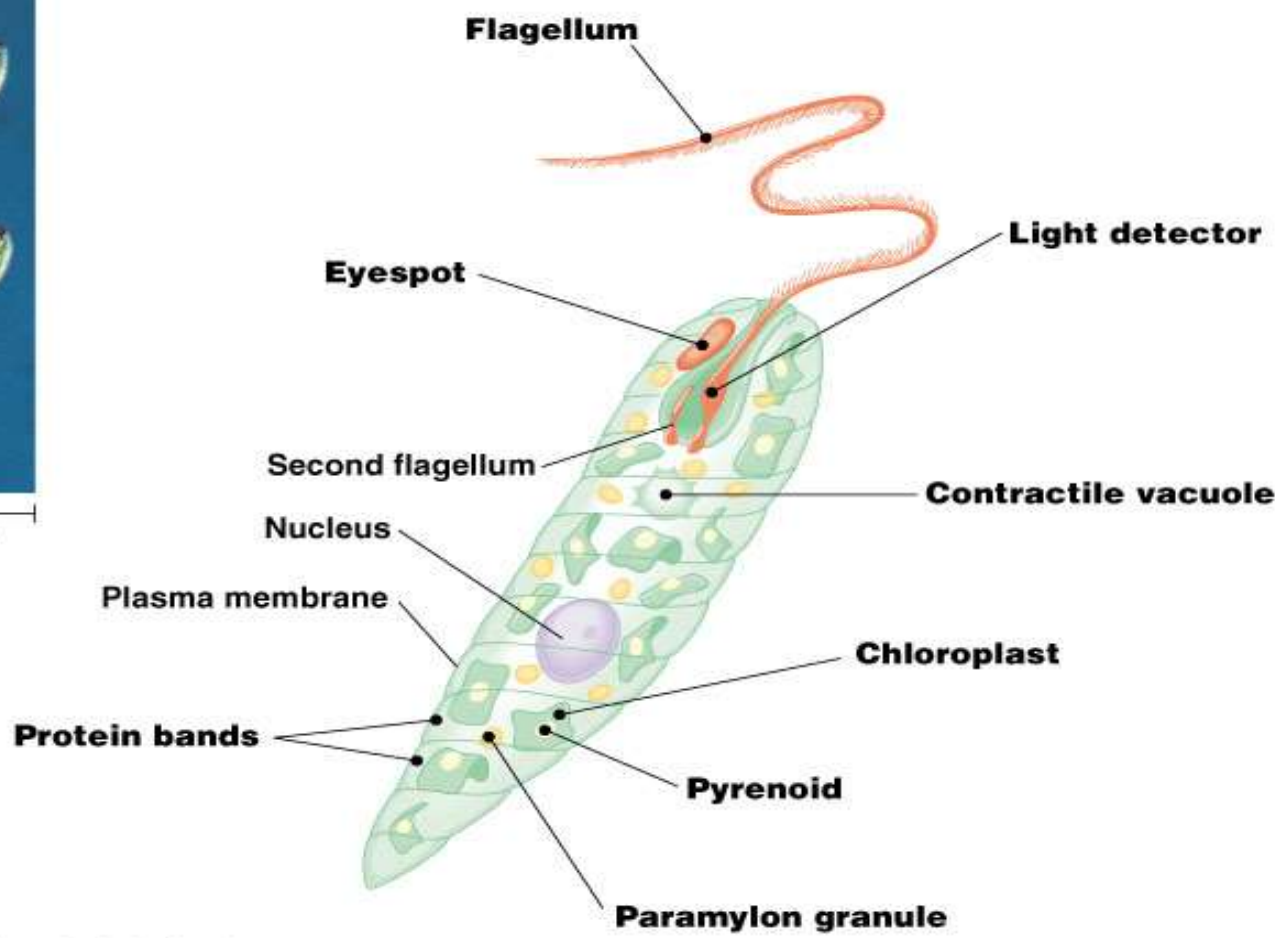
1. Euglenoids

Division: Euglenophyta



Euglena (LM)

0 µm



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Euglenoids

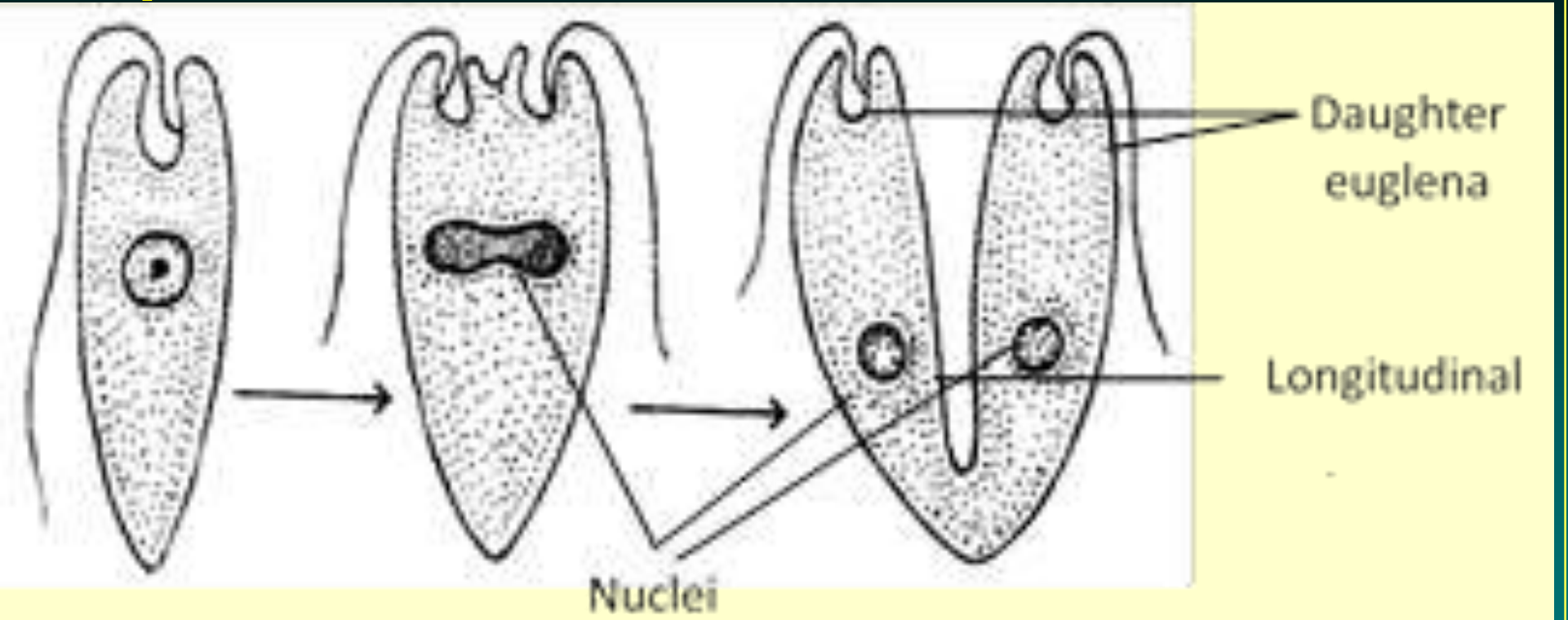
Euglena

- One-third of *euglenoids* have chloroplasts and in the abse nce of sunlight, they become heterotrophic.
- They have two flagella, one usually much longer than the other.
- No cell wall, motile
- A flexible *pellicle* allows euglenoids to assume different shapes (flexibility).
- Reproduction is by *longitudinal cell division*.

An euglena is a protist that can eat food like animals do (partly heterotrophic) and can make food like plants do (partly autotrophic). Euglena can surround a particle of food and eat it (phagocytosis). Many Euglena contain chloroplasts and chlorophyll a and b. Euglena live in fresh water, salt water and in marine-like environments. Many Euglena are able to move by using a flagellum, a long whip-like structure. When the water dries up, a euglena forms a protective wall around itself and lies dormant in the form of a spore until the environment improves. It reproduces by mitosis, splitting into parts, like amoeba.

Why Euglena sometimes can be considered to have both plant and animal features?!!!.

Reproduction



Longitudinal binary fission in euglena

•Asexual reproduction by longitudinal cell division starting from the front end of the cell

•Sexual reproduction: unknown

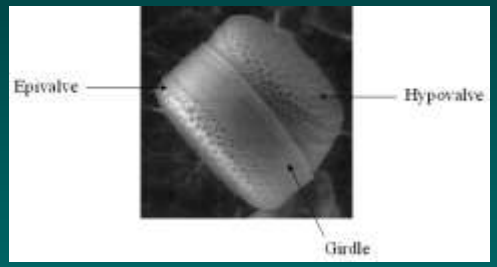
•**Cysts** are formed to survive unfavorable conditions; thick, mucilaginous cell wall, loss of flagella, eye-spot mostly present, rounding of cells, increase in paramylon granules

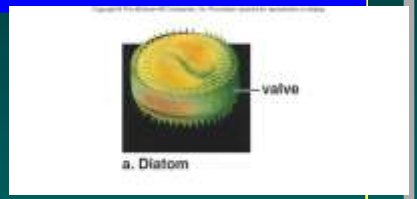
•**Germinating cysts**

2. Diatoms

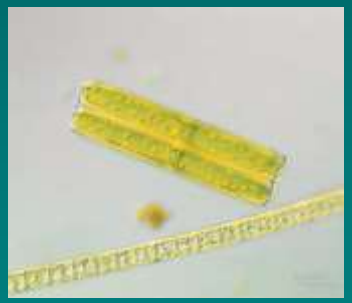
Division: Bacillariophyta

- beautiful one celled protists
- come in many shapes
  - EX: boats, rods, disks, triangles





- important food source for water dwelling animals
- cell covering is made up of 2 overlapping parts



- Think of a box with a lid

- made of the same material as glass "shells of silica"

- cell coverings do not decay (when dead)

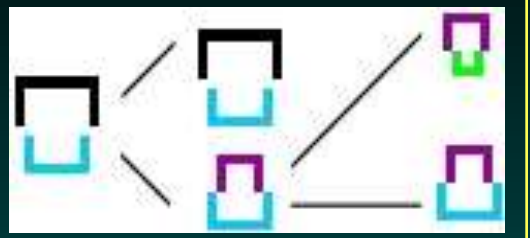
- used for toothpaste, scouring powders, & filters, detergents and as a component of dynamite. As it is also heat-resist ant, it can be used as a thermal insulator عازل حرارى

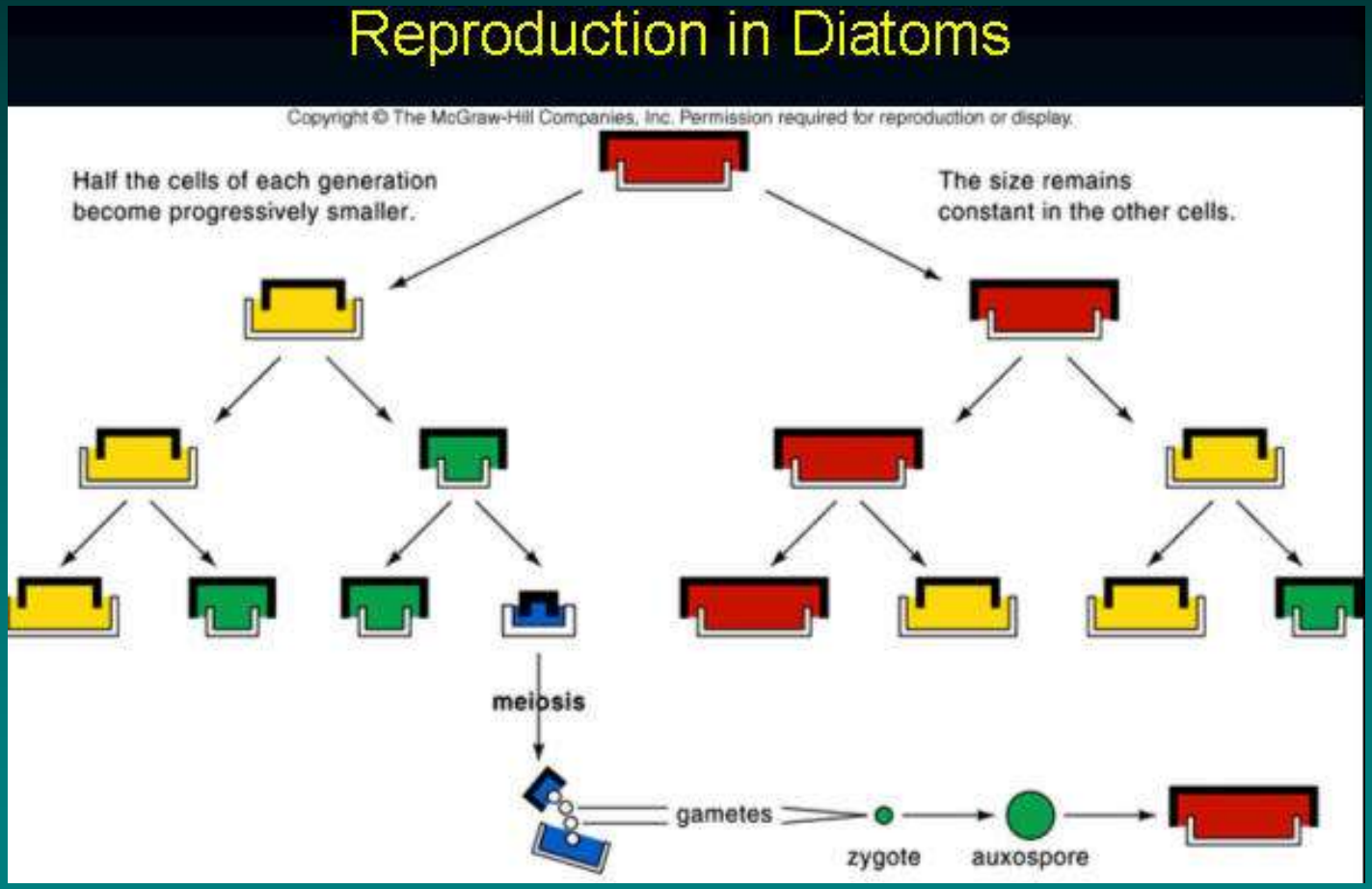
- **Diatomaceous earth consists of fossilized remains of diatoms, (a type of hard-shelled algae).**

Diatomaceous earth is mined from ancient deposits of marine di atoms. All that remains of the cells is the silica frustules, but th ese are extremely useful in a number of commercial ways such as abrasives and filters.

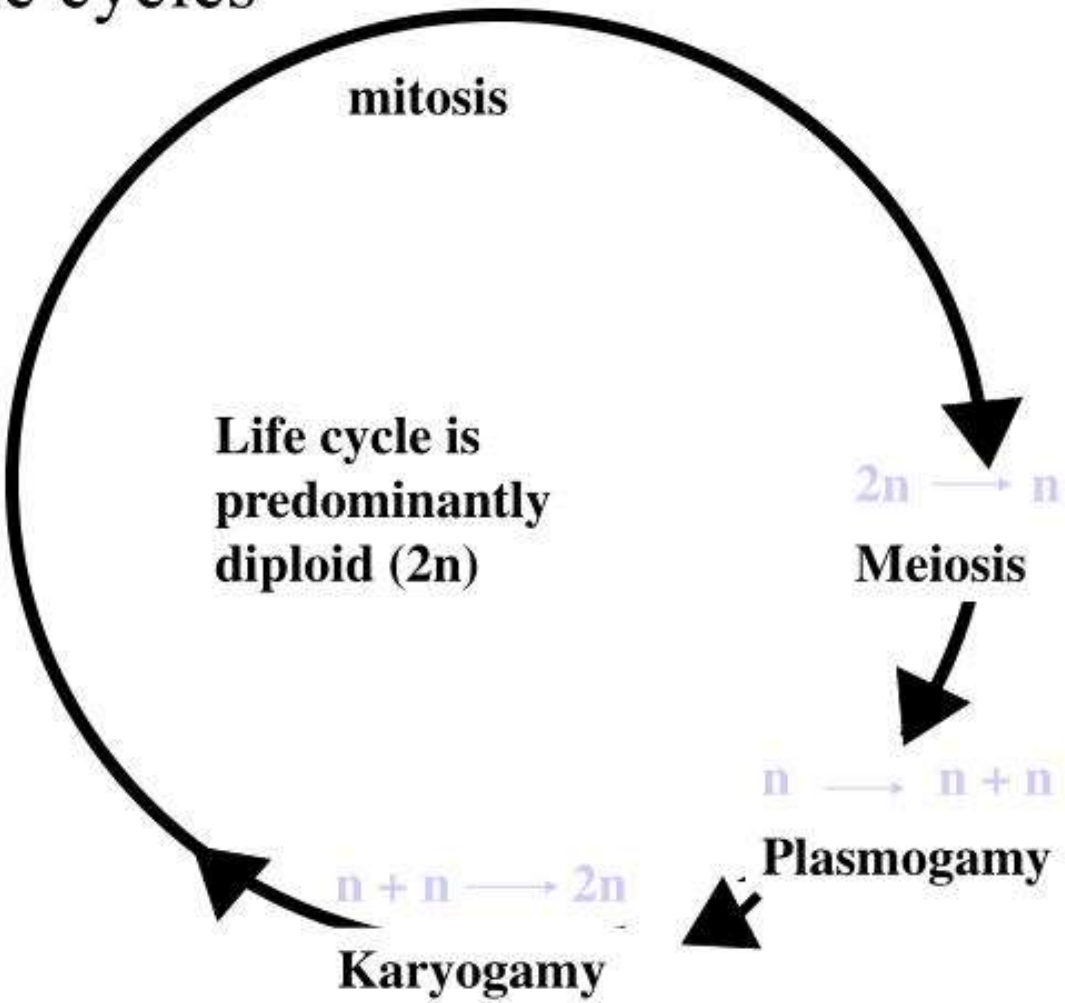
Photosynthetic pigment called carotenoids – give them a golden color

Asexual Reproduction





Fungal life cycles



Sexual Reproduction

•Differentiation of parent cells into gametangia
Next state is called gametogenesis: each cell divides meiotically forming spherical gametes.

•Each gametangium yields two morphologically isogamous nonflagellated gametes with aniso gamous behavior.

•One gametangium produces two active gametes while the other produces two passive gamete s.

•The frustules completely open, allowing both active gametes to move via amoeboid action to wards the passive gametes in the other gametangium.

•Each gamete fuses with the nearest one in the other gametangium, forming a spherical zygote in 1-2 minutes.

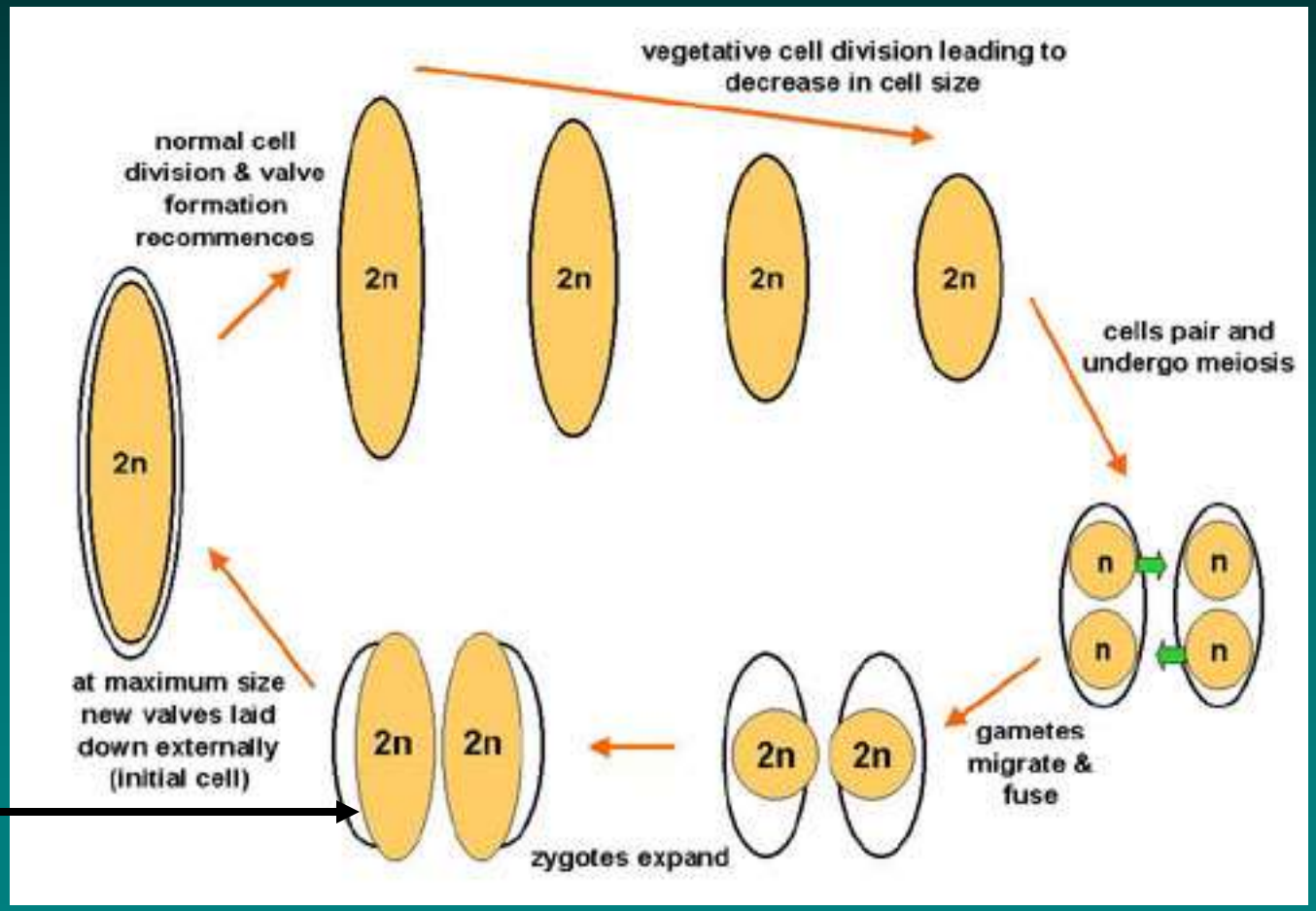
•Gametes were unable to fuse if they had lost contact with their parent frustule

•Two auxospores are formed from the two sets of gametes.

•They remain attached to only one parent frustule (or may not be associated to any frustule).

•Gamete formation to appearance of initial cells (auxospores) takes 2-4 days (entire process of sexual reproduction).

Diatoms auxospores:



Auxospore

3. "Red Tide" Dinoflagellates

Division: Dinoflagellata

- The third unicellular algae, Dinoflagellata, are a major component of marine phytoplankton.

- These algae have at least two flagella set at right angles to each other and thick cell walls made of cellulose plates.

- Blooms of dinoflagellates cause "Red Tide."

- Produce chemicals that kill thousands of fish
- Humans can become ill if they eat shell fi sh that have absorbed these chemicals
- Common off the coast of Florida or other warmer areas

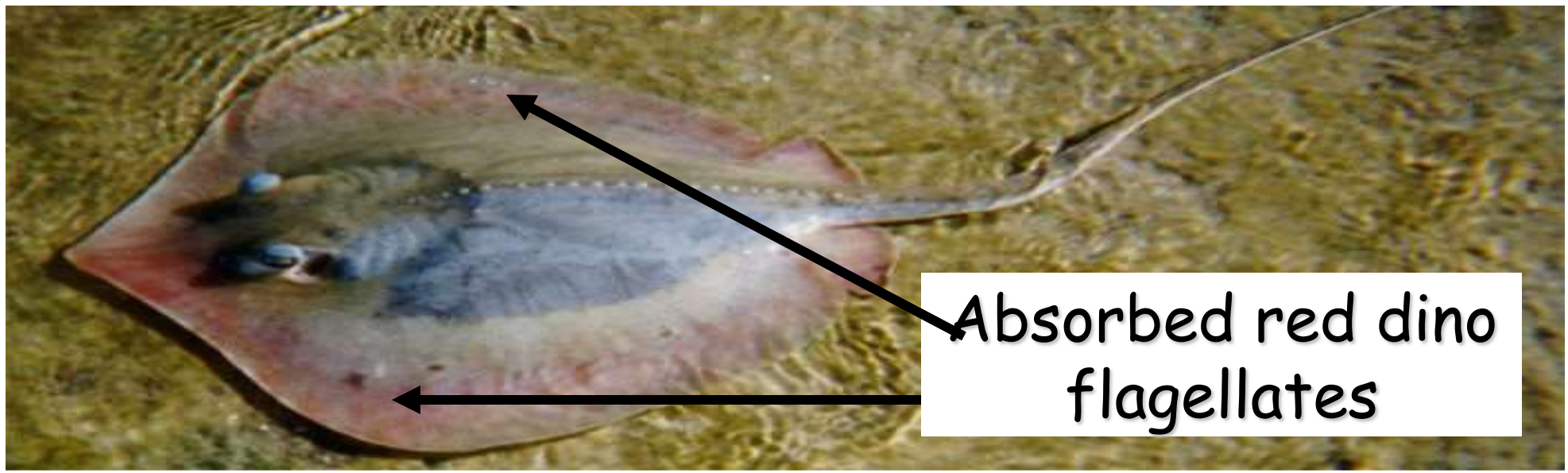
(Produce large numbers wh ich turns the ocean red)

المد الأحمـــــر

☹ Red Tide ☹

Red Tide in Kat O, Hong Kong (January 2001)

☹ Red Tide Kills Fish ☹



After encountering and absorbing a red tide..a stingr ay becomes disoriented. They have the appearance of not being able to see or control what they are doing. Eventually the ray will stay at the surface lying horiz ontally and die.

Pfiesteria-related fish lesions. Photo courtesy of the Aquatic Botany Lab, North Carolina State University.

They are the cause of "*red tides*" that kill fish because they contain poisons.

أفهم من كلامك ده أنك بتهددني ؟

طيب مش شارب الحليب وريني هتعمل ايه

Multicellular Plant-Like Protists

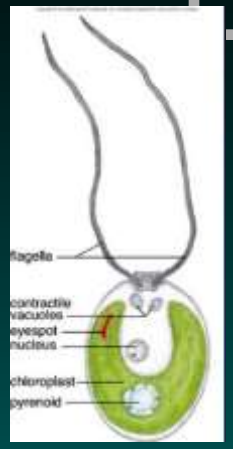
- Multicellular algae are classified by color.

Ex. Green Algae-Chlorophyta

- Most live in fresh water
- Can be unicellular or multicellular
- Live alone or in groups called colon ies

Like plants, *green algae* have a cell wall of cellulose, have both chlorop hylls *a* and *b*, and store reserve food as starch (pyrenoids).

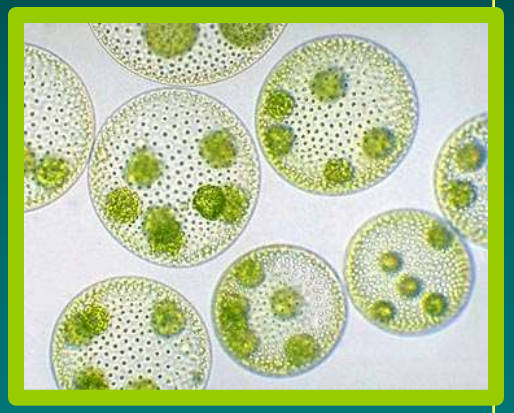
- Come in many different forms



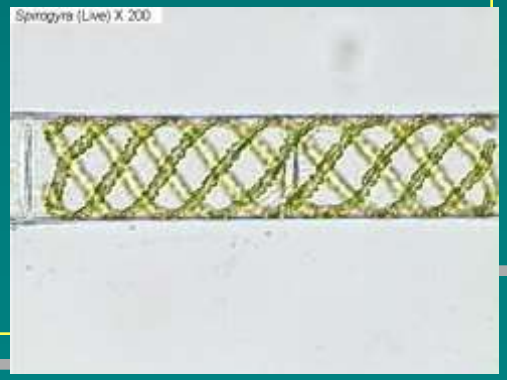
- Could be unicellular like *Chlamydomonas*

- Usually found in "colonies"

(Colony – a group of cells that live together)



Colonial green algae include *Volvox*, in which thousands of flagellated cell s are arranged in a single layer surrou nding a watery interior.

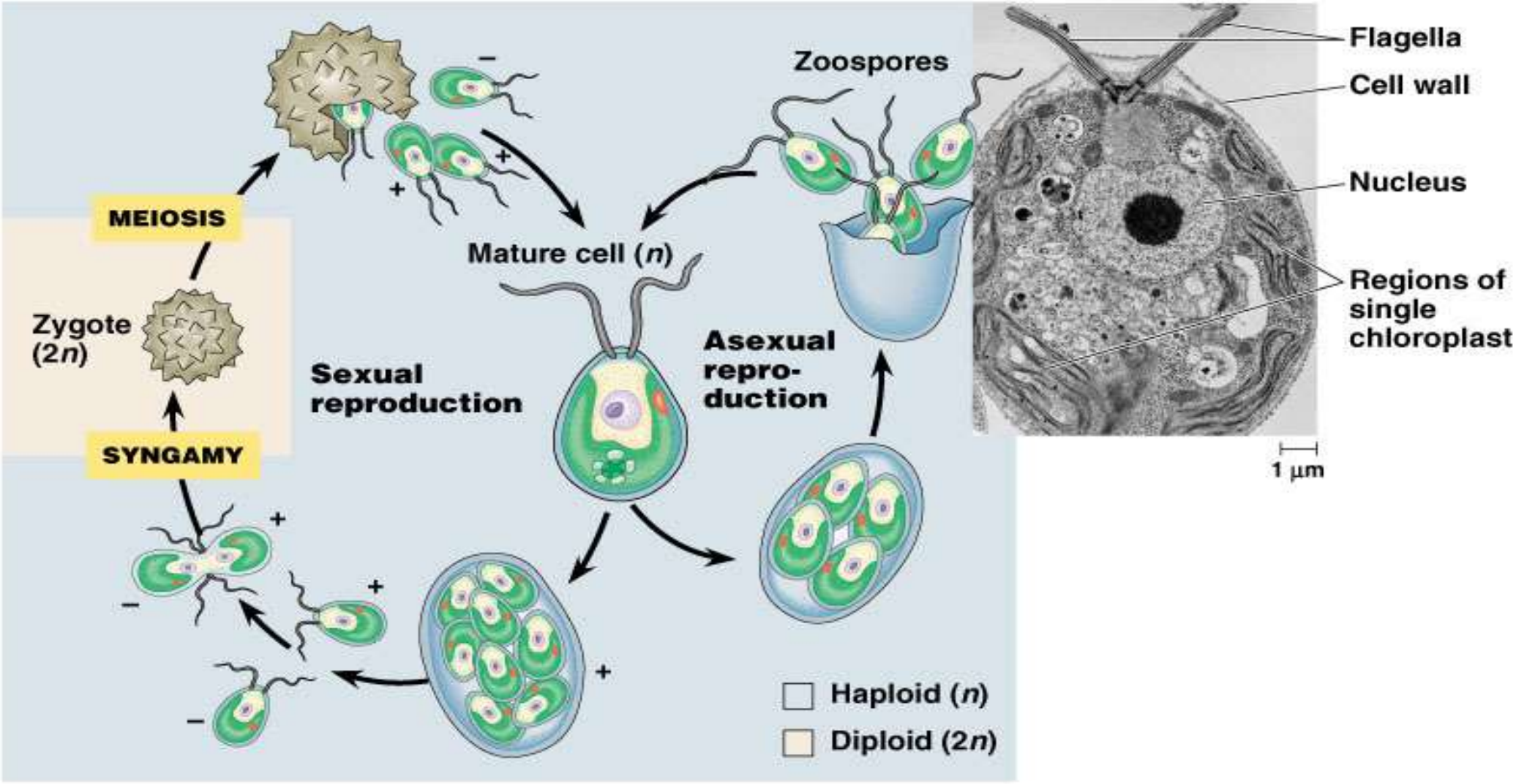


Filamentous green algae include *Sp irogyra* with long chains of cells tha t have spiral chloroplasts.

Multi-cellular chloro phyta (Chara-like sp ecies) evolved to be come the higher pla nts.

Chara- very closely r elated to first land pl ants

The life cycle of *Chlamydomonas*



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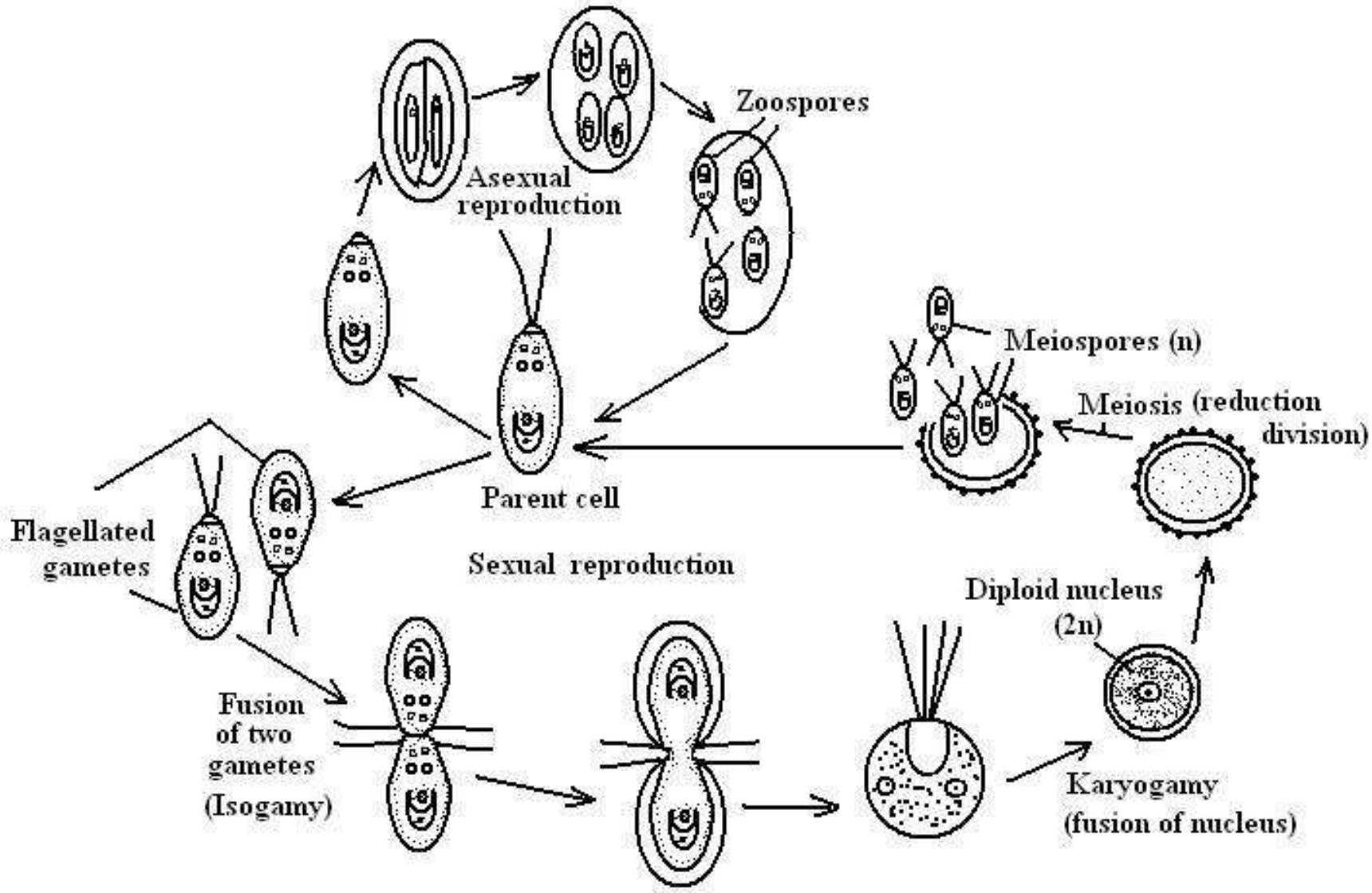


Fig. 11 Life cycle of Chlamydomonas

Life Cycle of *Chlamydomonas*

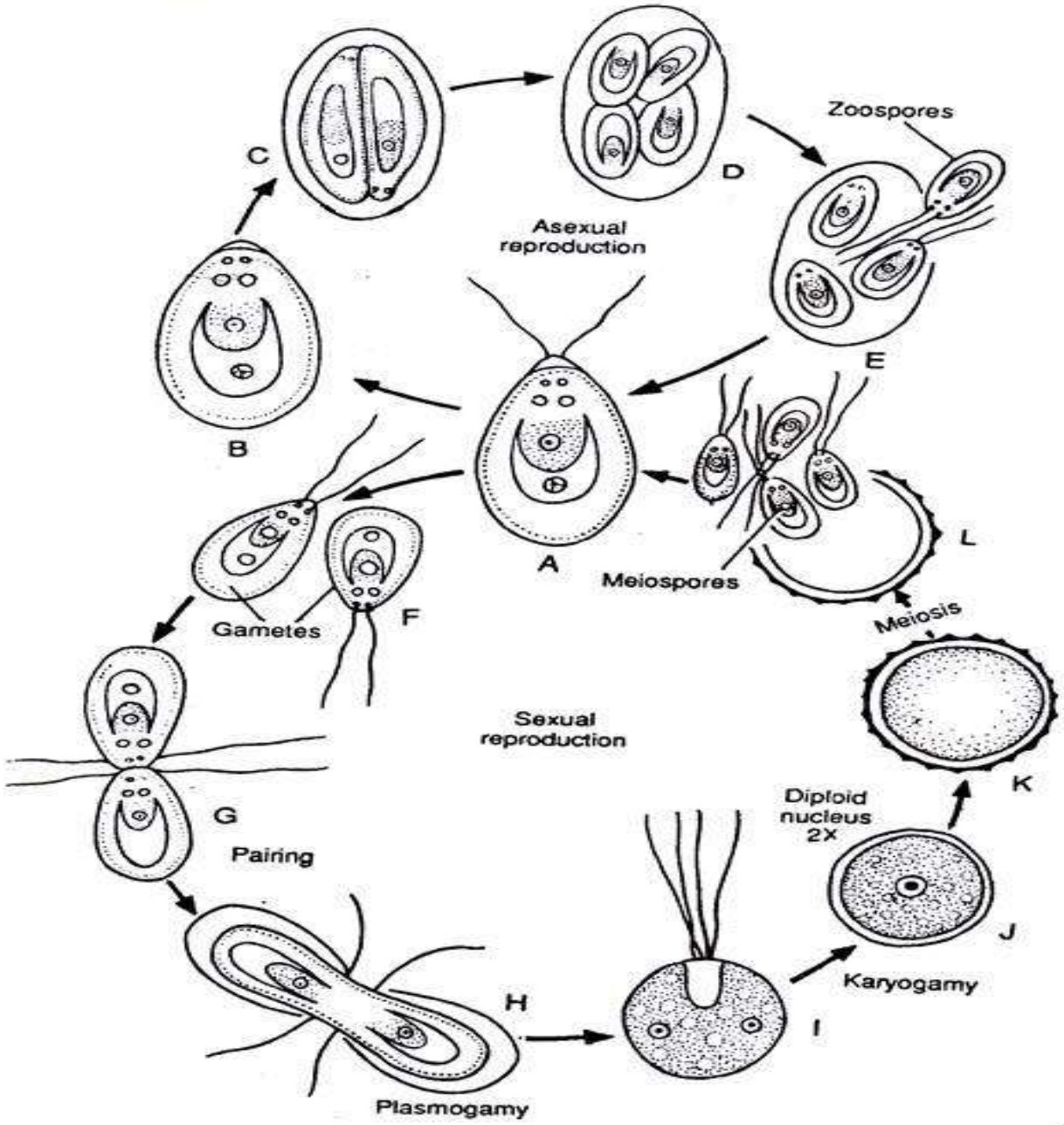
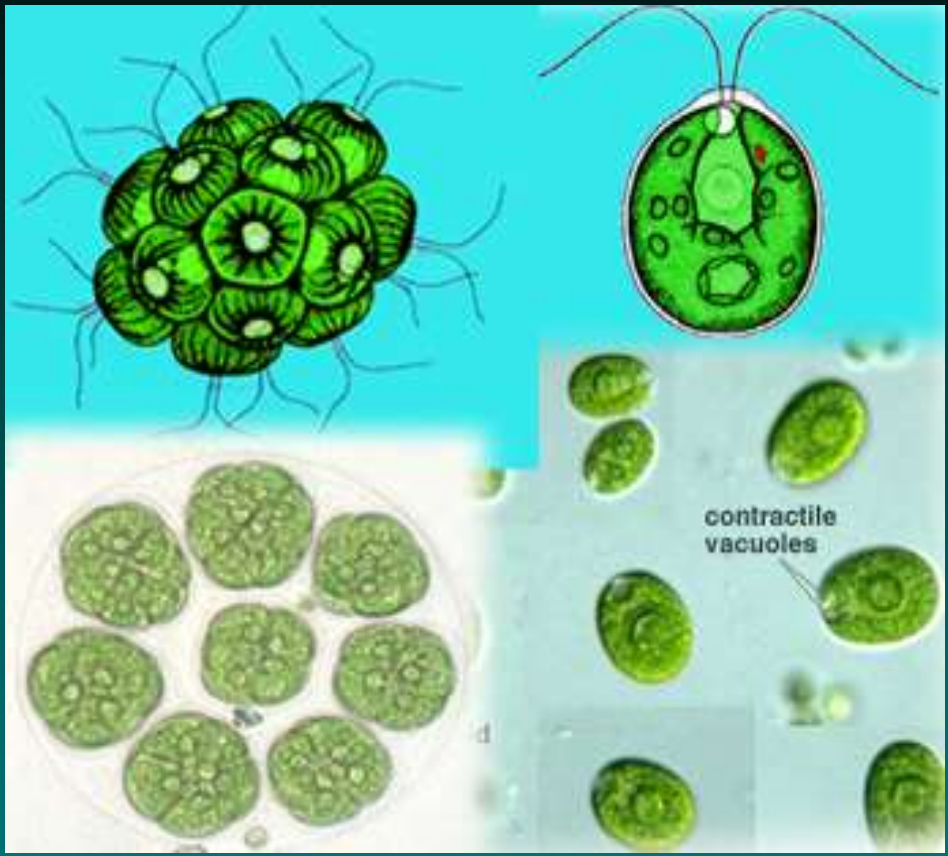


Fig. 14. *Chlamydomonas*. Diagrammatic life cycle

Look at this FLASH to understand sexual and asexual r eproduction of *Chlamydomonas* and answer the followi ng question

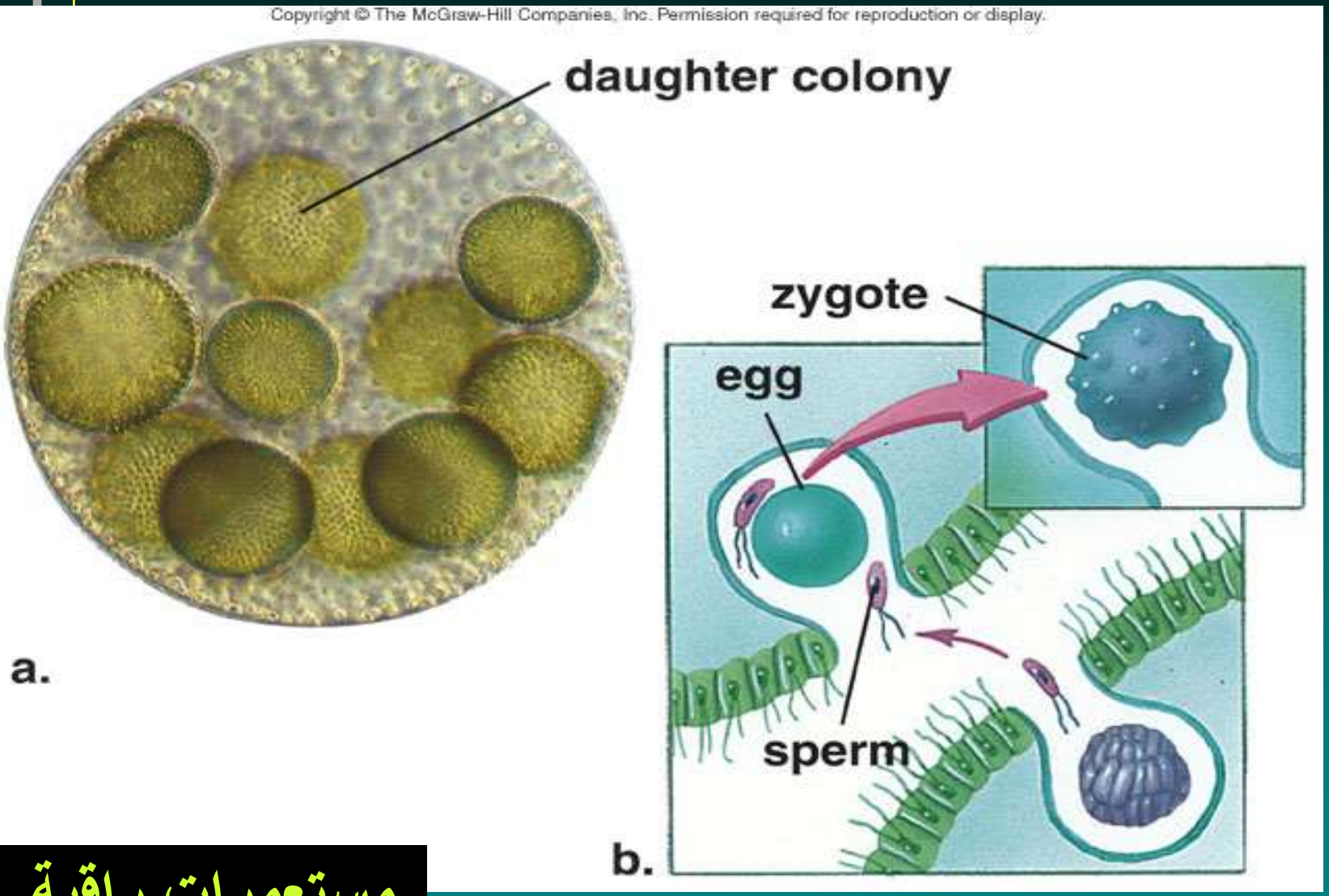
Which part of *Chlamydomonas* is dipl oid during its life cycle?



Pandorina & Chlamydomonas

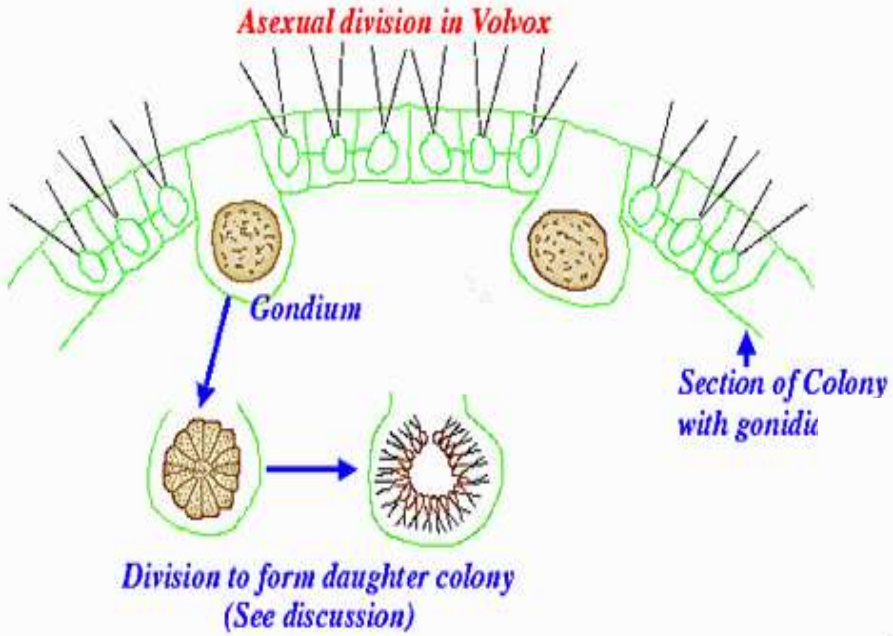
مستعمرات غير راقية

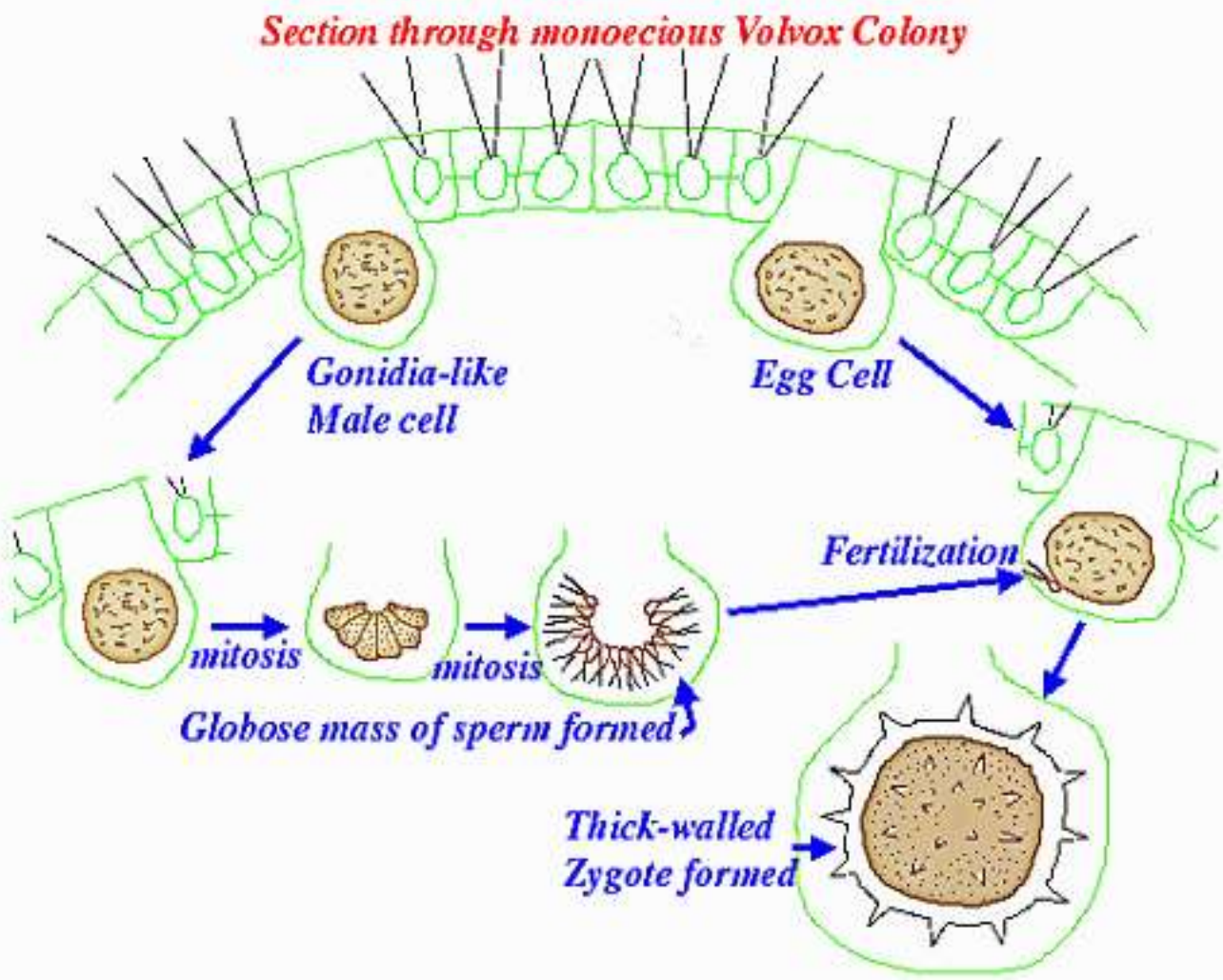
Volvox, a colonial green alga



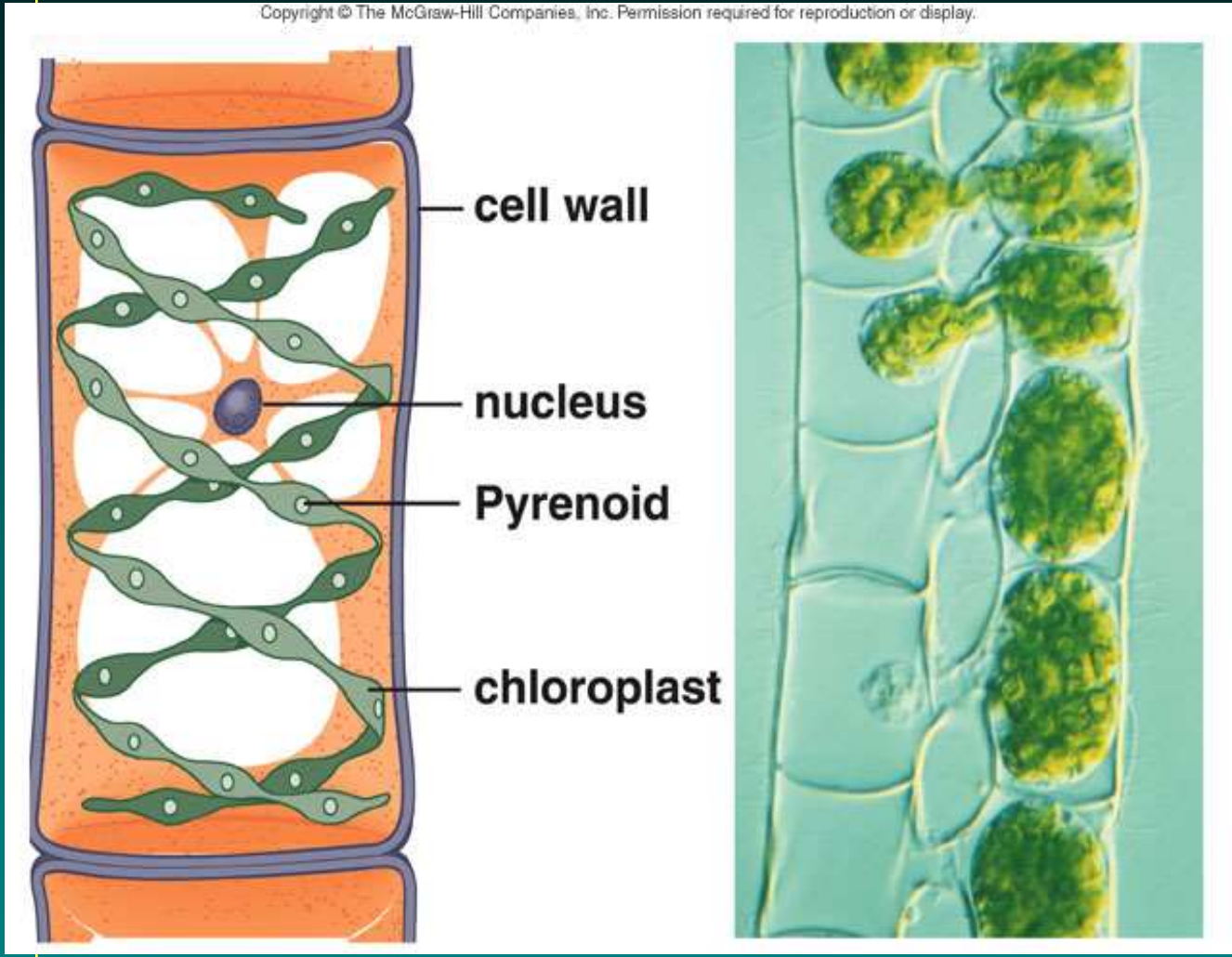
Colonies of unicellular organisms

مستعمرات راقية

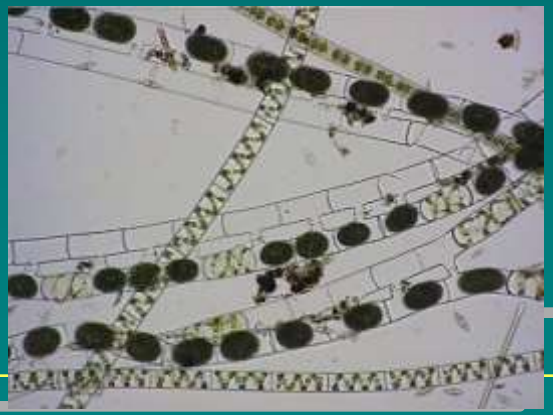

Asexual division in Volvox
Gondium
Section of Colony
with gonidia
Division to form daughter colony
(See discussion)


Section through monoecious Volvox Colony
Gonidia-like
Male cell
Egg Cell
Fertilization
mitosis
mitosis
Globose mass of sperm formed
Thick-walled
Zygote formed

Spirogyra, a filamentous green alga



- During *conjugation*, cel l contents of one filame nt move across to cells of other filament, formi ng diploid zygotes that s urvive the winter.
- The zygotes undergo m eiosis to produce new h aploid adult filaments.



Sclariform Reproduction

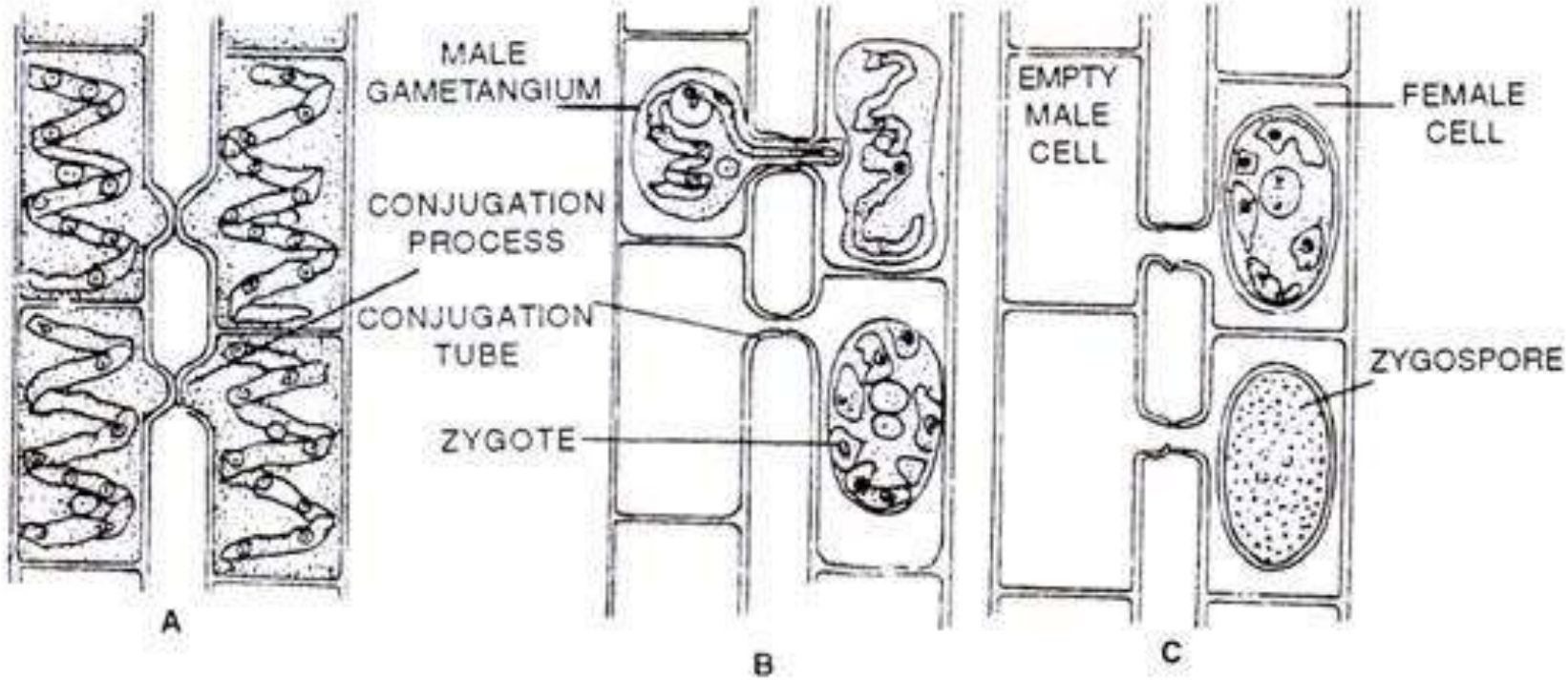
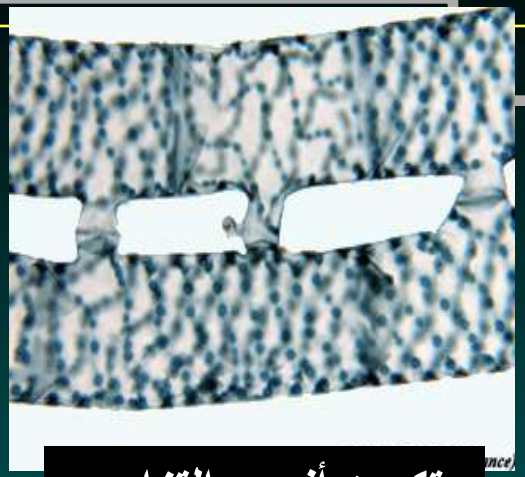


Figure 5.7 Stages in scalariform conjugation of *Spirogyra*.

Spirogera Sexual Reproduction

مراحل التكاثر الجنسى فى سبيروجيرا



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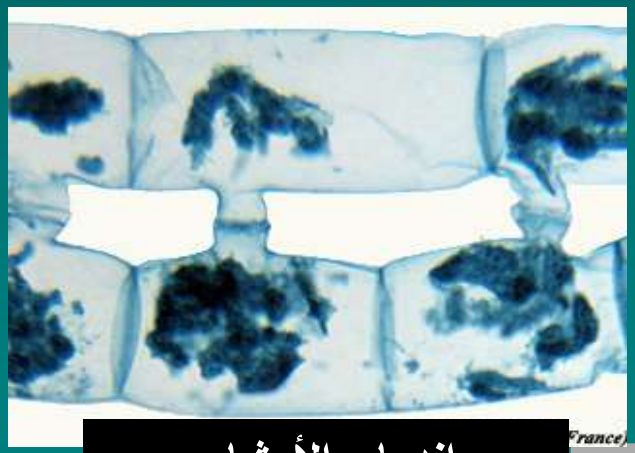

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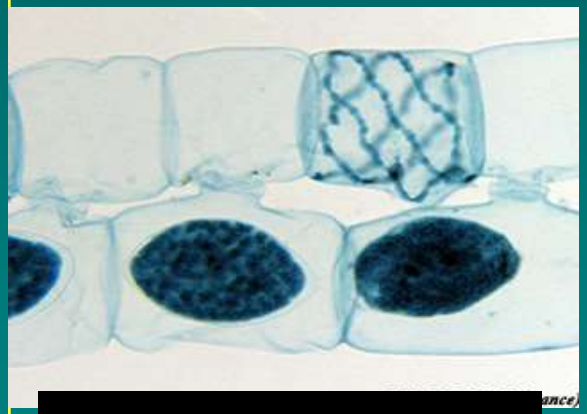
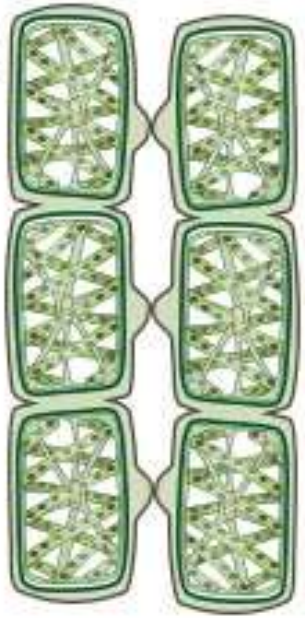
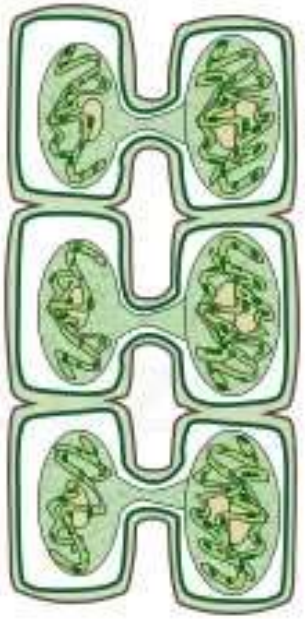


تكوين Zygospore

1
Two filaments come side by side

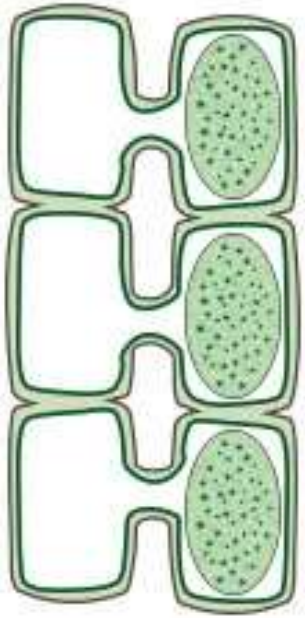
2
Conjugation tubes develop





3
Conjugation canals form

4
Male gametes fuse with female gametes to form zygotes



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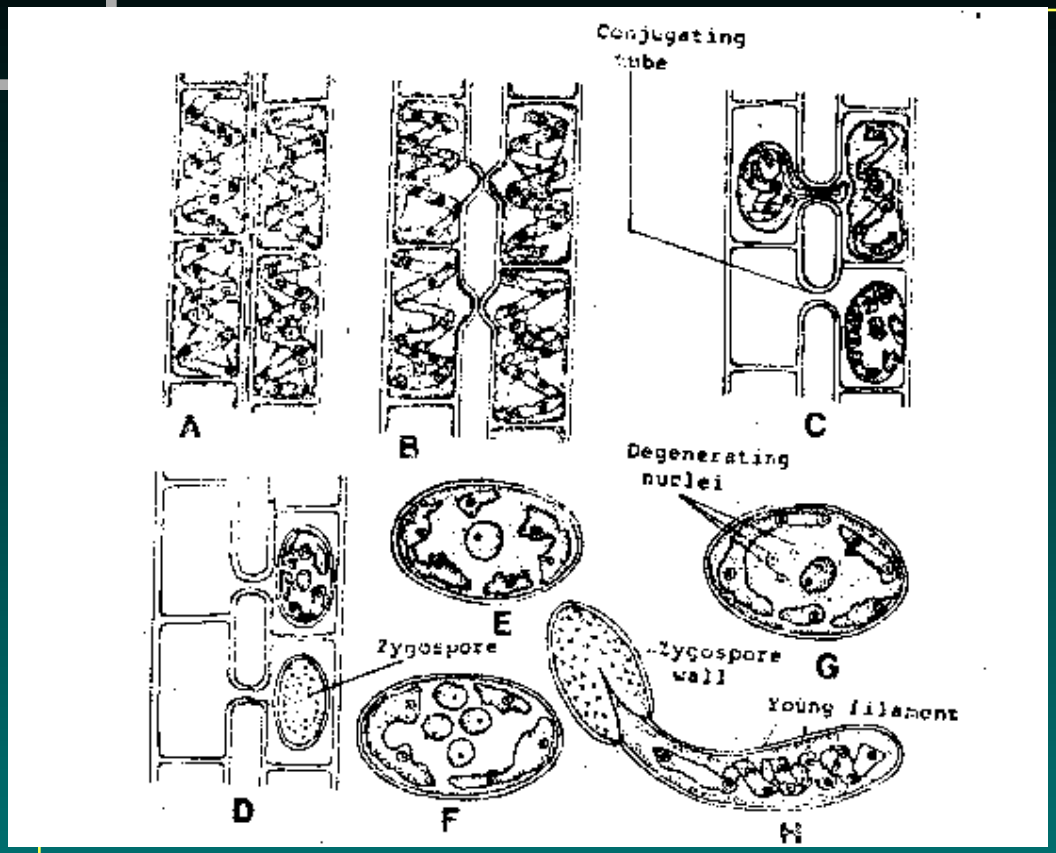
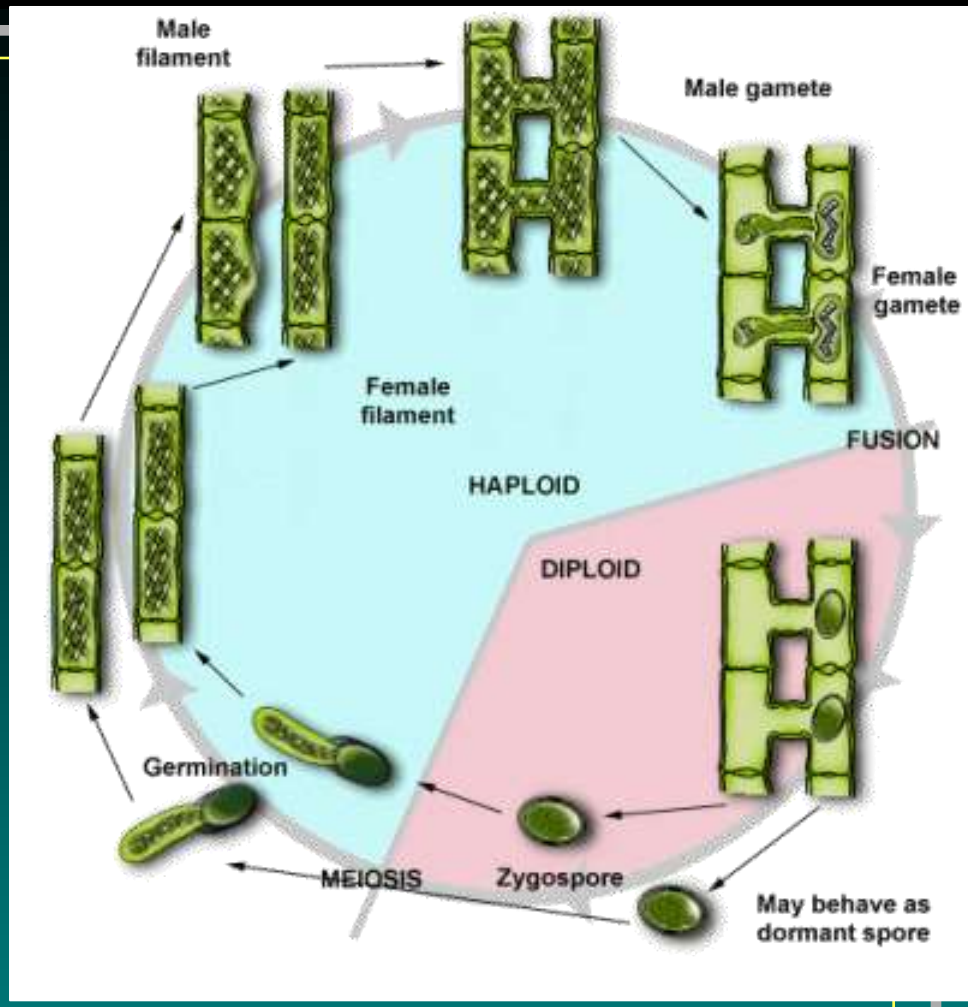


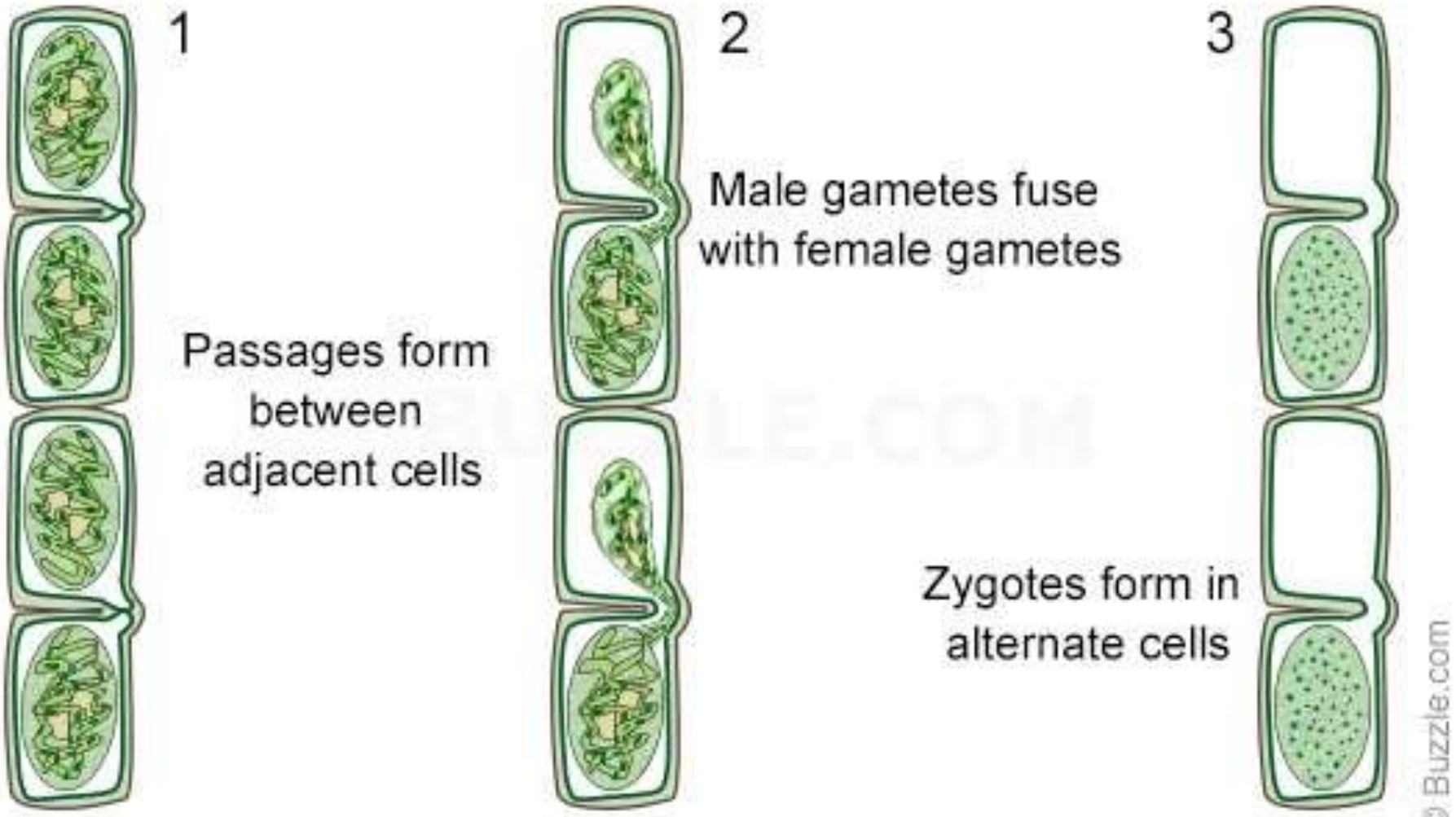


Spirogera Sexual Reproduction

مراحل التكاثر الجنسى فى سبيروجيرا

Lateral Sexual Reproduction

مراحل التكاثر
الجنسى الجانبى



و شكرا لحسن الاستماع